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Lab 6

[IT2120] Probability & Statistics

1. An IT company claims that their newly developed learning platform improves student performance in online tests. According to previous data, 85% of students who used the platform passed their online tests. A batch of 50 students is selected at random who have completed the course using this platform. Let X denote the number of students who passed the test out of 50 students.

   I. What is the distribution of X?

      - **Binomial(n = 50, p = 0.85)**

   II. What is the probability that at least 47 students passed the test?

```
> n <- 50
> p <- 0.85
> # ii. Probability that at least 47 students passed
> prob_at_least_47 <- 1 - pbinom(46, size = n, prob = p)
> round(prob_at_least_47, 4)
[1] 0.046
```

2. A call center receives an average of 12 customer calls per hour.

   I. What is the random variable (X) for the problem?

      - **Number of customer calls received in an hour**

   II. What is the distribution of X?

      - **Poisson(lambda = 12)**

   III. What is the probability that exactly 15 calls are received in an hour?

```
> prob_exactly_15 <- dpois(15, lambda)
> #Probability that exactly 15 calls are received
> round(prob_exactly_15, 4)
[1] 0.0724
```